UNIVERSITY OF EMBU

DEPARTMENT OF MATHEMATICS AND STATISTICS

**COURSE CODE:** SMA 446 **COURSE TITLE:** Second-Order Partial Differential Equations

**SEMESTER:** II **ACADEMIC YEAR:** 2025/2026

## COURSE PURPOSE

This course provides a systematic introduction to partial differential equations (PDEs) with emphasis on linear first- and second-order equations and their applications in physics, engineering, and biology. The course focuses on the formulation, classification, and analytical solution of classical PDEs, including the wave, heat, and Laplace equations. Solution techniques such as the method of characteristics, separation of variables, Fourier series, Fourier transforms, and special functions (Bessel and Legendre functions) are developed and applied to initial- and boundary-value problems on bounded and unbounded domains.

## EXPECTED LEARNING OUTCOMES:

By the end of this course you should be able to:

1. Formulate mathematical models using partial differential equations.
2. Solve first-order PDEs using the method of characteristics.
3. Classify linear second-order PDEs into elliptic, parabolic, and hyperbolic types.
4. Solve classical PDEs arising in physical and biological systems.
5. Apply separation of variables and eigenvalue methods to boundary-value problems.
6. Use Fourier series and Fourier transforms to obtain analytical solutions.
7. Solve PDEs involving Bessel and Legendre functions.
8. Interpret mathematical solutions in applied contexts.

## COURSE CONTENT AND OUTLINE:

| WEEK / DATE | TOPIC / SUB-TOPICS |
|---|---|
| Week 1: 14/01/2026 | Introduction to Partial Differential Equations |
| Week 2: 21/01/2026 | First-Order Partial Differential Equations |
| Week 3: 28/01/2026 | Classification of Second-Order Linear PDEs |
| Week 4: 04/02/2026 | The One-Dimensional Wave Equation |

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| Week 5: 11/02/2026 | CAT 1 |
| Week 6: 18/02/2026 | The Heat (Diffusion) Equation |
| Week 7: 25/02/2026 | Laplace's Equation and Potential Theory |
| Week 8: 04/03/2026 | Separation of Variables and Eigenvalue Problems |
| Week 9: 11/03/2026 | CAT 2 |
| Week 10: 18/03/2026 | Fourier Series Methods |
| Week 11: 25/03/2026 | Special Functions in PDEs |
| Week 12: 01/04/2026 | Fourier Transform Methods |

## Course Evaluation

CATs & Assignments: 30%
End of Semester Examination: 70%
**Total:** **100%**

## Core Reading Materials


1. Mohamed, H. (2021). Partial Differential Equations. S.L.: Lap Lambert Academic Publ.
2. Buchanan, J. R., & Shao, Z. (2018). A first course in partial differential equations. Singapore ; Hackensack, NJ : World Scientific Publishing Co
3. Shah, N. (2020). Partial Differential Equations: An Introduction. S.L.: CRC Press.
4. Tolstykh, V. A. (2020). Partial differential equations: An unhurried introduction. Berlin ; Boston : Walter de Gruyter
5. Henner, V., Belozerova, T., & Nepomniashchii, A. A. (2020). Partial differential equations: Analytical models and applications. Boca Raton, Florida : CRC Press, Taylor & Francis Group


## Recommended Reading Materials


1. Jost, J. (2013). Partial differential equations. New York : Springer
2. DuChateau, P., & Zachmann, D. W. (2011). Partial differential equations. New York: McGraw-Hill
3. Cioranescu, D., Donato, P., & Roque, M. P. (2018). An introduction to second order partial differential equations: Classical and variational solutions. Singapore; Hackensack, NJ: World Scientific Publishing


**Name of Lecturer:** Dr. Dominic Kitavi **Sign:**  **Date:** 02-01-2026